# HEALTHCARE PATIENT DATA ANALYSIS

Business Case Study | Python Exploratory Data Analysis

**Project domain:** Healthcare Analytics
**Project type:** Exploratory Data Analysis (EDA)
**Author:** Shrii
**Tools:** Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

**A practical analysis of patient demographics, medical conditions, admission patterns, billing information and hospital stay duration.**

# 1. Executive Summary

Healthcare organizations generate large volumes of patient information, including demographic details, medical conditions, admission records, treatment information and billing data. Exploring these records can help reveal patterns in patient characteristics and hospital operations.

This project applies Python-based Exploratory Data Analysis to a healthcare dataset containing 55,500 initial records and 15 columns. Pandas and NumPy were used for data handling and numerical operations, while Matplotlib and Seaborn were used for visual exploration.

The work covers data understanding, quality checks, duplicate removal, date conversion, feature engineering, univariate, bivariate and multivariate analysis, correlation analysis and IQR-based outlier investigation.

**Important: Findings in this case study describe patterns in the supplied dataset. They should not be generalized to all hospitals or treated as causal or clinical conclusions.**

# 2. Business Problem

A healthcare organization wants to better understand its patient records, including demographics, medical conditions, admission patterns, billing and length of stay. Before drawing conclusions, the dataset needs to be inspected and prepared so that obvious data-quality issues do not undermine the analysis.

The project addresses questions such as:

- How are patients distributed across genders and age groups?
- Which medical conditions and admission types are most frequently recorded?
- How do billing amounts vary across medical conditions and admission types?
- Is patient age associated with billing amount?
- Do hospital stay durations differ across admission types or patient groups?
- What relationships appear among numerical variables, and are there unusual observations?

**Business objective:** Explore the dataset, identify meaningful patterns and produce evidence-based observations about patient records and hospital operations.

# 3. Project Objectives

- Understand the structure and characteristics of the dataset.
- Check missing values and duplicate records.
- Clean and prepare date fields and create an analytical Length of Stay feature.
- Explore patient demographics, medical conditions and admission patterns.
- Examine billing amounts and hospital stay duration.
- Study relationships between two or more variables.
- Use correlation and IQR methods to investigate numerical relationships and potential outliers.
- Summarize findings, limitations and areas for further investigation.

# 4. Dataset Description

The dataset initially contained 55,500 records and 15 columns. The columns include:

| Column | Description |
|---|---|
| Name | Patient name |

| Column | Description |
|---|---|
| Age | Patient age |
| Gender | Patient gender |
| Blood Type | Patient blood group |
| Medical Condition | Recorded medical condition |
| Date of Admission | Hospital admission date |
| Doctor | Treating doctor |
| Hospital | Hospital name |
| Insurance Provider | Insurance company |
| Billing Amount | Recorded hospital billing amount |
| Room Number | Assigned room number |
| Admission Type | Elective, Urgent or Emergency |
| Discharge Date | Patient discharge date |
| Medication | Medication recorded |
| Test Results | Patient test result category |

| Dataset metric | Value |
|---|---|
| Initial records | 55,500 |
| Columns | 15 |
| Exact duplicate rows identified | 534 |
| Records after duplicate removal | 54,966 |
| Missing values detected in initial check | 0 |

# 5. Tools and Technologies

| Tool | Purpose |
|---|---|
| Python | Programming and analysis |
| Pandas | Data manipulation, cleaning, grouping and summaries |
| NumPy | Numerical operations |
| Matplotlib | Data visualization |
| Seaborn | Statistical visualization |
| Jupyter Notebook | Interactive development environment |

# 6. Project Methodology

### Stage 1 — Data Loading and Understanding

The CSV file was imported using Pandas. The notebook used functions such as **head()**, **tail()**, **shape**, **columns**, **info()**, **describe()**, **nunique()** and **dtypes**.

These functions were used to inspect sample records, dimensions, column names, data types, non-null counts, descriptive statistics and the number of unique values. The initial inspection showed 55,500 records and 15 columns, with no missing values indicated by the non-null counts.

### Stage 2 — Data Quality Checks and Cleaning

### A. Missing-value analysis

The notebook used **df.isnull().sum()** to count missing values in each column. The initial check showed zero missing values across all 15 columns. Therefore, no imputation was performed on the basis of this check.

### B. Duplicate-record analysis

The notebook used **df.duplicated().sum()** and identified 534 exact duplicate rows. These were removed using **df.drop_duplicates()**. The resulting dataset contained 54,966 rows and 15 columns.

| Measure | Records |
|---|---|
| Initial dataset | 55,500 |
| Exact duplicates removed | 534 |
| Dataset after removal | 54,966 |

The duplicate share was approximately 0.96% of the original records. Duplicate removal was limited to fully identical rows; this does not establish whether there are other kinds of repeated patient records.

### C. Category and unique-value inspection

The notebook also used **nunique()** and **value_counts()** to inspect the number and frequency of categories in fields such as gender, medical condition, admission type, medication and test results.

### Stage 3 — Date Preprocessing and Feature Engineering

The Date of Admission and Discharge Date columns were converted from their original string/object representation to datetime using **pd.to_datetime()**. The **errors='coerce'** option converts unparseable values to missing datetime values, which should be checked after conversion.

**Date conversion should be followed by validation of newly created missing dates and of discharge dates occurring before admission dates. These checks are important before relying on Length of Stay.**

A new numerical feature, **Length of Stay**, was created by subtracting Date of Admission from Discharge Date and extracting the difference in days:

```
df['Length of Stay'] = (df['Discharge Date'] - df['Date of Admission']).dt.days
```

This feature supports analysis of average stay duration and comparisons across admission types, medical conditions and other patient groups.

## 7. Exploratory Data Analysis

### Stage 4 — Univariate Analysis

Univariate analysis examines one variable at a time to understand its distribution, frequency and characteristics. The project explored the following:

| Analysis | Question and method |
|---|---|
| Gender distribution | Compared the number of patients by gender using value_counts() and a countplot. |
| Age distribution | Used a histogram with KDE and descriptive statistics to examine patient ages. |
| Medical conditions | Used value_counts() and a countplot to compare the six recorded condition categories. |
| Admission types | Compared Elective, Urgent and Emergency admissions using counts and a countplot. |
| Billing amount | Used a histogram with KDE to examine the distribution and spread of billing values. |
| Length of Stay | Used a histogram with KDE to examine hospital stay duration. |
| Test results | Used value_counts() and a countplot to examine test-result categories. |

**Recorded visual observations:** Male and female patient counts appeared nearly equal. The age distribution covered a broad range and appeared relatively even across many intervals. The six medical-condition categories and three admission categories also appeared to have relatively similar frequencies. These are visual observations; exact counts and percentages should be taken from the notebook output.

## Stage 5 — Bivariate Analysis

Bivariate analysis examines the relationship between two variables. Five comparisons were explored:

| Comparison | Method and interpretation |
|---|---|
| Gender vs Length of Stay | Grouped average stay by gender and displayed a barplot. The graph showed approximately 15.5 days for male p |
| Medical Condition vs Billing | Grouped billing by medical condition and compared average values in a barplot. Obesity appeared highest and C |
| Admission Type vs Billing | Summarized mean, median, minimum and maximum billing by admission type and used a boxplot to compare dis |
| Age vs Billing | Used a scatterplot and Pearson correlation to examine the direction and strength of a linear association. The exa |
| Admission Type vs Length of Stay | Compared mean, median, minimum and maximum stay by admission type and used a boxplot to examine distribu |

A barplot compares group summaries, while a boxplot displays the median, interquartile range and potential extreme observations. A scatterplot helps reveal patterns between two numerical variables.

## Stage 6 — Multivariate Analysis

Multivariate analysis examines three or more variables together. Two analyses were performed:

### A. Medical Condition, Gender and Admission Type

A Pandas **crosstab()** grouped Medical Condition and Gender against Admission Type to show counts across the three categorical variables. A countplot compared medical conditions by admission type. The displayed chart suggested relatively balanced admission-type frequencies across conditions. The visual chart itself did not separately encode gender; gender detail was available in the cross-tabulation.

The visual observations recorded during the project included comparatively higher Urgent admissions for Diabetes, Emergency admissions for Obesity, and Elective admissions for Hypertension and Cancer. These should be checked against the cross-tabulation before being treated as exact rankings.

### B. Medical Condition, Admission Type and Billing Amount

A Seaborn barplot used Medical Condition on the x-axis, Billing Amount on the y-axis and Admission Type as the hue. This compared average billing across combinations of condition and admission type. The chart is useful for examining group differences, but does not by itself explain why any differences occur.

### Stage 7 — Correlation Analysis

Numerical columns were selected using **select_dtypes(include=np.number)**. Room Number was dropped because it is an identifier, not a meaningful continuous measurement. A correlation matrix was calculated using **.corr()** and visualized using a Seaborn heatmap with annotated coefficients.

Correlation coefficients range from -1 to +1. Positive values indicate a positive linear association, negative values indicate a negative linear association, and values near zero indicate little linear association. Correlation does not establish causation. Exact coefficients and any interpretation should be copied from the actual heatmap.

### Stage 8 — Outlier Analysis

Potential outliers were investigated in Billing Amount and Length of Stay using boxplots and the Interquartile Range (IQR) method.

The method calculates Q1 (25th percentile), Q3 (75th percentile) and IQR = Q3 – Q1. The conventional lower and upper fences are Q1 – 1.5 × IQR and Q3 + 1.5 × IQR. Values outside these fences are flagged as potential outliers.

The project generated boxplots for both variables and created filtered records for values outside the IQR fences. Exact outlier counts must be taken from the executed notebook; a boxplot screenshot alone is not sufficient to confirm that the count is zero.

**Important data-quality note: the initial descriptive statistics showed a minimum Billing Amount of approximately -2,008.49. Negative billing values require separate investigation (for example, refund, adjustment or data-entry issue). They are not automatically invalid and are not necessarily IQR outliers. Do not remove them without understanding their meaning.**

Outliers were not automatically removed because unusually high bills or long stays may represent genuine observations.

## 8. Business Findings and Interpretation

The work produced the following supported visual observations and analytical directions:

- Male and female patient counts appeared nearly equal in the gender visualization.
- The dataset represented a broad range of patient ages, with relatively similar frequencies across many displayed age intervals.
- The six medical-condition categories had relatively similar frequencies in the countplot.
- Elective, Urgent and Emergency admissions appeared relatively balanced in the admission-type visualization.
- Average Length of Stay appeared nearly identical for male and female patients (approximately 15.5 versus 15.4 days in the displayed chart).
- Obesity appeared to have the highest average billing and Cancer the lowest among the six medical-condition categories, although the differences appeared modest.
- The cross-tabulation and grouped charts provide ways to compare admission patterns and billing across patient categories.
- Negative billing values were observed and need separate validation before final billing conclusions are treated as reliable.

The exact metric outputs for average billing by admission type, average stay by medical condition, medication frequencies, test-result frequencies, correlation coefficients and IQR outlier counts should be inserted from the notebook before publication. They are not inferred here.

## 9. Business Recommendations

- **Strengthen data-quality checks:** Monitor duplicate records, invalid dates, missing values and unusual numerical entries as part of routine data preparation.

- **Review negative billing entries:** Confirm whether negative amounts represent refunds, credits, billing adjustments or errors before using billing averages for decisions.
- **Monitor Length of Stay:** Use the engineered feature to compare stay duration across admission types and medical conditions, while checking date validity.
- **Investigate billing differences:** If group-level billing differences persist after validation, examine additional fields such as procedures, treatment intensity, resource use and insurance coverage.
- **Maintain consistent data definitions:** Document units, date conventions, category labels and the meaning of billing values to support reliable future analysis.

# 10. Limitations

- The analysis is limited to the columns and records available in the dataset.
- The dataset's distributions may not represent the patient population of a real hospital.
- The available fields do not establish the reasons behind billing differences or length-of-stay patterns.
- Correlation describes association, not causation.
- Negative billing entries and date validity require additional validation.
- The analysis is exploratory and does not provide clinical recommendations.

# 11. Conclusion

This project demonstrated a structured Python-based EDA workflow applied to healthcare patient records. The work began with dataset inspection and data-quality checks, identified and removed 534 exact duplicate rows, converted admission and discharge dates to datetime, and created a Length of Stay feature.

Univariate analysis explored individual distributions. Bivariate and multivariate analyses examined relationships among demographics, medical conditions, admission types, billing and hospital stay. Correlation analysis was used to examine linear relationships between numerical variables, while boxplots and IQR fences were used to investigate potential outliers.

The project strengthened practical skills in data cleaning, preprocessing, feature engineering, statistical exploration, visualization and communicating analytical observations. The findings should be interpreted within the limitations of the dataset and validated before being used for operational decisions.

# 12. Skills Demonstrated

| Technical skills | Analytical skills |
|---|---|
| Python and Jupyter Notebook | Business-question formulation |
| Pandas and NumPy | Pattern identification |
| Matplotlib and Seaborn | Data interpretation |
| Data cleaning and preprocessing | Critical thinking |
| Feature engineering | Analytical storytelling |
| GroupBy, value_counts and crosstab | Evidence-based observations |
| Correlation and IQR analysis | Communicating limitations |

# 13. Project Information

| Field | Details |
|---|---|
| Project title | Healthcare Patient Data Analysis – Python EDA |
| Repository name | Healthcare-Patient-Data-Analysis-Python-EDA |
| GitHub profile | https://github.com/Shrii888 |
| Author | Shrii |
| Project domain | Healthcare Analytics |
| Project type | Exploratory Data Analysis |
| Tools | Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook |

Suggested repository: https://github.com/Shrii888/Healthcare-Patient-Data-Analysis-Python-EDA